

Dynamic optimization of condition-based maintenance strategies using Monte Carlo simulations to compare adaptive inspection intervals, threshold adjustments, and their impact on system performance

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ABSTRACT

Condition-based maintenance (CBM) has emerged as a pivotal strategy for enhancing operational efficiency and minimizing maintenance costs in complex systems. This paper presents a comparative analysis of three adaptive maintenance strategies: the fixed-interval policy ($\Delta T, M$), the adaptive-interval policy ($\Delta T_k, M$), and the adaptive-threshold policy ($\Delta T, M_k$). The performance of each strategy is evaluated in terms of its impact on maintenance costs, system downtime, and reliability, using a stochastic degradation model based on the Wiener process. A Monte Carlo simulation framework is employed to simulate the degradation process and maintenance actions across a predefined operational horizon. The results demonstrate that the adaptive strategies consistently outperform the fixed-interval policy, reducing both maintenance costs and downtime while ensuring higher system reliability. Furthermore, penalties for reliability shortfall and excessive between-inspection risk are introduced to provide a comprehensive evaluation of strategy performance. This study contributes to the field of adaptive maintenance by highlighting the advantages of dynamic maintenance scheduling and threshold adjustment in managing operational risks and optimizing system performance.

1. Introduction

Condition-based maintenance (CBM) has become a widely implemented strategy in industries seeking to optimize operational efficiency and reduce maintenance costs [1]. Unlike traditional time-based maintenance, which schedules inspections and maintenance at fixed intervals regardless of the system's condition, CBM aims to perform maintenance actions based on the actual condition of system components. This approach minimizes unnecessary maintenance interventions and reduces downtime, resulting in a more cost-effective and reliable operation [2]. Advancements in CBM have been facilitated by the integration of stochastic degradation models, which enable more accurate predictions of the degradation process of system components [3–6]. These models provide essential information for estimating the remaining useful life (RUL) of components, which is used to determine the optimal time for preventive maintenance (PM) and corrective maintenance (CM) [7–10]. Among the most widely used models for system degradation is the Wiener process, a stochastic model that effectively captures the continuous deterioration of components over time, allowing for the

prediction of system failure under uncertain conditions [11]. Despite the advantages offered by CBM, challenges remain in selecting the optimal maintenance strategy. In systems characterized by high variability and unpredictable degradation, dynamic adjustment of inspection intervals and maintenance thresholds is necessary [12–17]. Adaptive maintenance strategies, which adjust these parameters based on the system's real-time condition, have emerged as a potential solution to these challenges. These strategies aim to improve system performance by more effectively aligning maintenance actions with the actual degradation state of the system, thus optimizing both cost and reliability [18–22].

This study evaluates and compares the performance of three adaptive maintenance strategies: the fixed-interval policy ($\Delta T, M$), the adaptive-interval policy ($\Delta T_k, M$), and the adaptive-threshold policy ($\Delta T, M_k$). The objective of the study is to assess the effectiveness of these strategies in terms of system reliability, maintenance cost, downtime, and the frequency of maintenance actions [23–25]. The evaluation process employs a Monte Carlo simulation framework, which simulates the degradation process and maintenance actions over a predefined

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operational horizon, providing a robust comparison of the strategies [26–30]. Additionally, the study incorporates penalties for reliability shortfall and excessive between-inspection risk. The penalty for reliability shortfall is applied when the system's reliability falls below a predefined threshold of $R_{\min}=0.95$ at the reporting horizon [31]. Similarly, a penalty is applied for excessive between-inspection risk, which quantifies the likelihood that the system's degradation exceeds the failure threshold between inspections [32]. These penalties serve to enhance the evaluation framework by providing a comprehensive measure of the strategies' performance in terms of both cost and risk management [33,34].

Recent advances in stochastic prognostics have further extended the capabilities of condition-based maintenance frameworks by incorporating uncertainty in degradation dynamics and failure thresholds. In particular, adaptive remaining useful life (RUL) prediction models with stochastic failure limits have been proposed to better capture variability in system behavior under complex operating conditions [35]. These approaches highlight the importance of explicitly modeling uncertainty in both degradation evolution and failure criteria [36,37].

In parallel, observer-based methods and fault estimation techniques have been developed to improve state estimation in nonlinear and partially observable systems. Such methods enable more accurate tracking of system degradation and can enhance maintenance decision-making by providing real-time information on system health [38,39].

Although the present study focuses on inspection-based adaptive maintenance strategies using a Wiener degradation model, these recent developments suggest that integrating predictive models, observer-based estimation, and real-time monitoring could significantly improve reliability performance. In particular, they offer promising directions for reducing between-inspection risk and enhancing the effectiveness of adaptive maintenance policies [40,41].

Despite the extensive use of stochastic degradation models and Monte Carlo simulation in condition-based maintenance, important gaps remain in the comparative evaluation of adaptive maintenance strategies under uncertainty. In particular, the ability of adaptive mechanisms to effectively control failure risk has not been fully addressed.

The main contribution of this study is to provide a unified framework for comparing fixed-interval, adaptive-interval, and adaptive-threshold maintenance policies within a consistent stochastic setting. In addition, a penalty-augmented objective function is introduced, incorporating both reliability shortfall and between-inspection risk, allowing for a more comprehensive assessment of maintenance performance.

The results reveal a key limitation of adaptive inspection-based strategies: while they improve cost efficiency and the allocation of preventive and corrective actions, they do not significantly reduce between-inspection failure risk. This finding highlights a fundamental gap in current condition-based maintenance approaches, where local adaptation mechanisms are insufficient to ensure global reliability performance under stochastic degradation.

The structure of this paper is as follows: Section 2 outlines the methodology employed in the study, including the stochastic degradation model, condition-based maintenance decision logic, and the simulation framework. Section 3 presents the Monte Carlo simulation architecture used to evaluate these strategies. Section 4 discusses the simulation configuration and input parameters, while Section 5 presents and interprets the results of the simulation study. Finally, Section 6 concludes the paper, summarizing the key findings and outlining potential avenues for future research.

2. Methodology

This section describes the methodology used to evaluate and compare the performance of three adaptive condition-based maintenance (CBM) strategies. The study employs a stochastic degradation model and Monte Carlo simulations to assess the effectiveness of the strategies in terms of system reliability, cost, downtime, and

maintenance actions. The maintenance strategies incorporate adaptive inspection scheduling and threshold adjustments based on system degradation.

2.1. Stochastic degradation model

The degradation process of the system is modeled as a Wiener process with drift, described by the stochastic differential equation (SDE) [42]:

$$dX(t) = \mu dt + \sigma dW(t), \quad (1a)$$

Where $X(t)$ represents the degradation state at time t , μ is the drift coefficient indicating the rate of degradation, σ is the diffusion coefficient representing random fluctuations in degradation, and $dW(t)$ is the increment of a standard Wiener process that introduces randomness into the degradation process [43] [44].

For a Wiener degradation process with drift, the time required for the degradation trajectory to reach the failure threshold L_f , referred to as the first-passage time τ , follows an inverse Gaussian distribution. The corresponding probability density function is given by:

$$f_\tau(t) = \frac{L_f}{\sqrt{2\pi\sigma^2 t^3}} \exp\left(-\frac{(L_f - \mu t)^2}{2\sigma^2 t}\right) \quad (1b)$$

and the cumulative distribution function is:

$$P(\tau \leq t) = \Phi\left(\frac{\mu t - L_f}{\sigma\sqrt{t}}\right) + \exp\left(\frac{2\mu L_f}{\sigma^2}\right) \Phi\left(-\frac{\mu t + L_f}{\sigma\sqrt{t}}\right) \quad (1c)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

These expressions characterize the boundary-crossing behavior of the degradation process and provide a theoretical foundation for estimating failure probabilities and remaining useful life (RUL). In this study, these probabilities are primarily evaluated using Monte Carlo simulation, which approximates the first-passage-time distribution through repeated sampling of stochastic degradation trajectories. This approach ensures flexibility in handling complex maintenance policies while remaining consistent with the underlying stochastic theory.

The system is considered to have failed when the degradation state $X(t)$ exceeds a predefined failure threshold L_f [34]:

$$X(t) \geq L_f, \quad (2)$$

At this point, corrective maintenance (CM) is initiated to restore the system to its initial condition. The system begins at $X(0)=0$, which represents the “as-good-as-new” state [45] [46].

2.2. Maintenance strategies

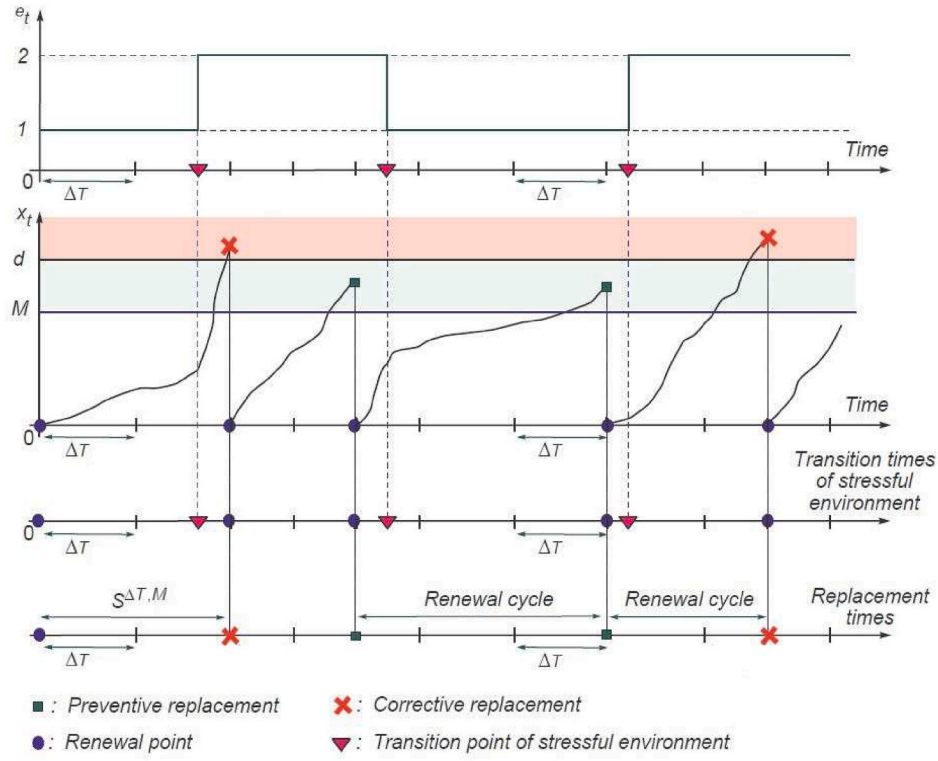
Three maintenance strategies are evaluated in this study: the fixed-interval policy ($\Delta T, M$), the adaptive-interval policy ($\Delta T_k, M$), and the adaptive-threshold policy ($\Delta T, M_k$) [41]. These strategies determine when preventive maintenance (PM) is performed and how inspection intervals or thresholds are adjusted based on system condition [47] [48].

2.2.1. Fixed-interval policy ($\Delta T, M$)

The fixed-interval policy schedules inspections at fixed time intervals, ΔT , regardless of the system's degradation state (see Figure (1)). Preventive maintenance is triggered when the degradation state $X(t)$ exceeds a fixed threshold L_m , as expressed by the condition [41]:

$$X(t) \geq L_m, \quad (3)$$

The inspection intervals remain constant, and the system's degradation does not influence the timing of the inspections. Thus, ΔT is independent of the system condition [49].

Fig. 1. Principle of maintenance strategy $(\Delta T, M)$.

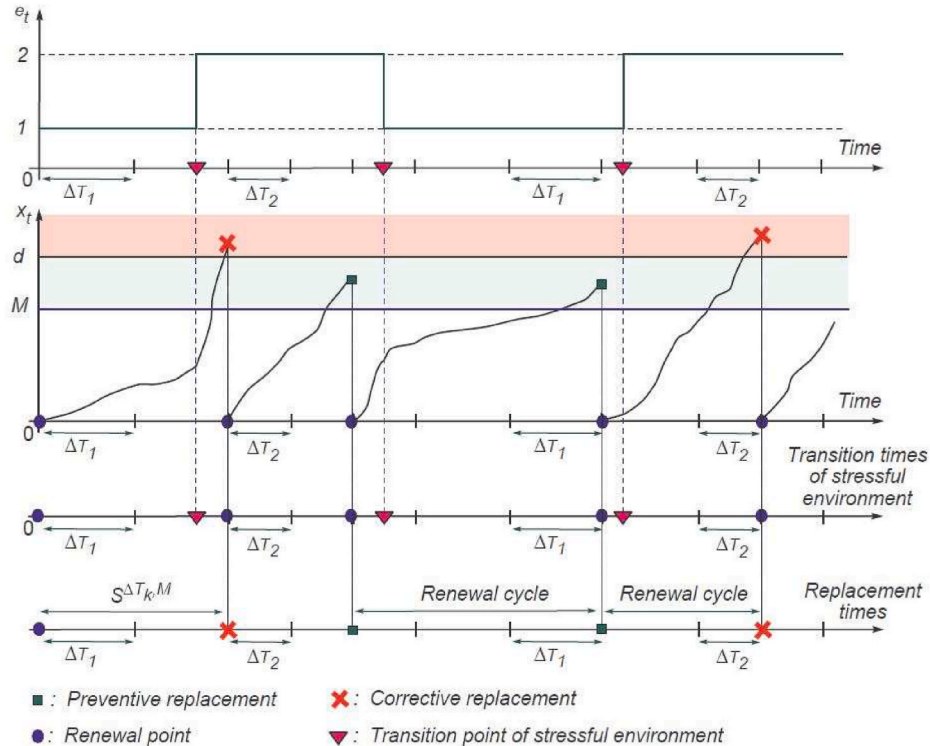
2.2.2. Adaptive-interval policy $(\Delta T_k, M)$

The adaptive-interval policy dynamically adjusts the inspection interval ΔT_k based on the system's degradation (see Figure (2)). The inspection interval is determined by the system's remaining useful life (RUL), calculated as the time remaining until the failure threshold L_f is

reached. The RUL at time t is defined as [50]:

$$RUL(t) = \frac{L_f - X(t)}{\mu}, \quad (4)$$

The inspection interval ΔT_k is then adapted according to the

Fig. 2. Principle of maintenance strategy $(\Delta T_k, M)$.

following equation [50]:

$$\Delta T_k = \min \left(\max \left(\alpha_T \frac{L_f - X(t)}{\mu}, \Delta T_{\min} \right), \Delta T_{\max} \right), \quad (5)$$

Where:

- α_T is the adaptation parameter that controls the rate of change in inspection intervals,
- $\Delta T_{\min}$ and $\Delta T_{\max}$ are the minimum and maximum allowable inspection intervals, respectively.

This policy aims to reduce inspection frequency when the system is in good condition and increase it when the system is near failure, thereby improving system reliability and reducing unnecessary maintenance costs [51].

2.2.3. Adaptive-threshold policy ($\Delta T, M_k$)

The adaptive-threshold policy modifies the threshold for preventive maintenance based on the system's RUL (see Figure (3)). The threshold $L_{m,k}$ is adjusted at each inspection epoch based on the degradation state $X(t)$. The threshold adjustment follows the equation [49]:

$$L_{m,k} = \min(L_f - \delta L_{\text{safety}}, \max(L_{\min}, L_m - \alpha_L (L_f - X(t)))) \quad (6)$$

Where:

- α_L is the adaptation parameter controlling the threshold adjustment rate,
- $L_{\min}$ is the minimum allowable threshold,
- δL_{safety} ensures that the threshold remains sufficiently below the failure limit to provide a safety margin.

Preventive maintenance is triggered when the degradation state exceeds the adjusted threshold $L_{m,k}$.

2.3. Cost model

The cost model includes the costs associated with preventive maintenance, corrective maintenance, and inspections. The total cost of a given simulation run is the sum of the costs associated with each maintenance action and inspection, as well as the downtime incurred.

- Preventive Maintenance (PM) Cost: Each preventive maintenance action incurs a cost C_{PM} , and the downtime associated with PM is t_{PM} .
- Corrective Maintenance (CM) Cost: Each corrective maintenance action incurs a cost C_{CM} , which includes repair costs and associated penalties. The downtime associated with corrective maintenance is t_{CM} .
- Inspection Cost: Each inspection incurs a cost C_{insp} , with a downtime of t_{insp} .

The total cost for a simulation run is computed by summing the costs of all preventive and corrective maintenance actions and inspections, along with their associated downtimes.

2.4. Reliability and risk

The reliability of each strategy is evaluated using the Kaplan–Meier survival estimate, which provides the survival probability $S(t)$ at a given time t . Reliability is determined by:

$$R(t) = S(t), \quad (7)$$

Where $S(t)$ is the probability that the system remains operational up to time t . The reliability requirement is set as $R_{\min} = 0.95$, and each strategy is assessed based on its ability to meet or exceed this threshold.

The between-inspection risk p_{hit} is the probability that the system's degradation exceeds the failure threshold L_f before the next inspection. This risk is given by Ref. [52]:

$$p_{hit} = 1 - S(t), \quad (8)$$

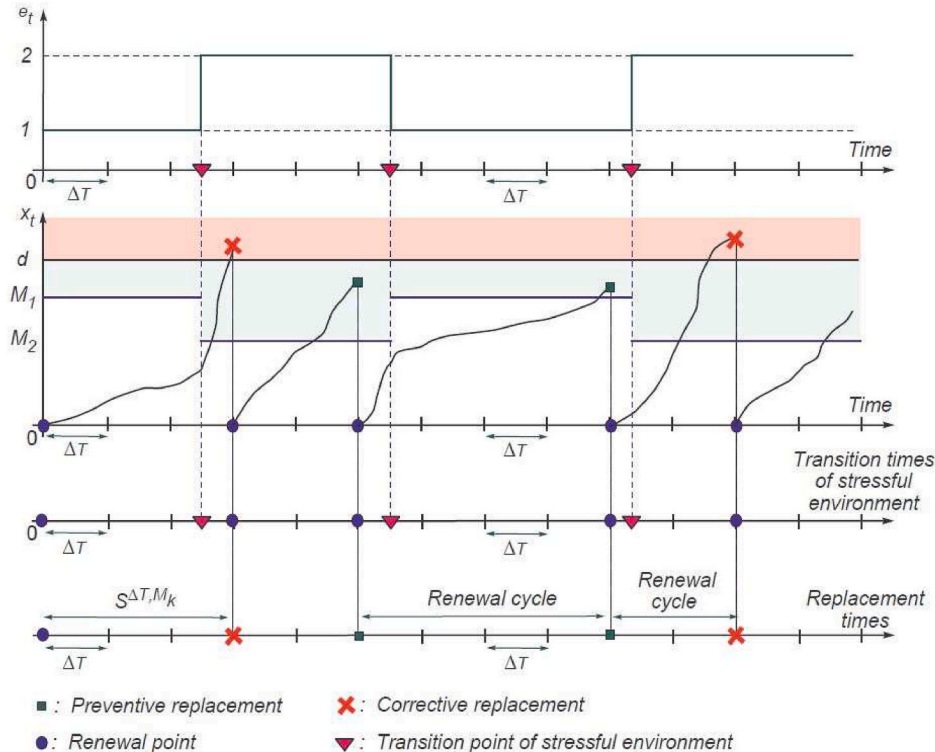


Fig. 3. Principle of maintenance strategy ($\Delta T, M_k$).

Where $S(t)$ is the survival function evaluated at the time of inspection. Minimizing this risk is one of the objectives of the maintenance strategies, ensuring the system operates without failure between inspections [53] [54].

2.5. Penalty for reliability shortfall

A penalty is applied for any shortfall in reliability below the required threshold $R_{\min}$. The penalty is calculated as the difference between the required reliability and the observed reliability at the reporting horizon t_{report} [55]:

$$\text{Penalty}_R = \lambda_R \cdot \max(0, R_{\min} - R(t_{\text{report}})), \quad (9)$$

Where λ_R is the penalty weight for the reliability shortfall. This penalty is included in the penalty-augmented objective function, which evaluates the overall cost of each maintenance strategy.

2.6. Penalty for excessive between-inspection risk

In addition to the reliability penalty, a penalty is imposed for excessive between-inspection risk. The penalty for risk p_{hit} for each inspection interval is calculated as [55]:

$$\text{Penalty}_{\text{hit}} = \lambda_{\text{hit}} \cdot \frac{1}{N_{\text{inspections}}} \sum_{i=1}^{N_{\text{inspections}}} p_{\text{hit}}(i), \quad (10)$$

Where $N_{\text{inspections}}$ is the total number of inspections during the simulation period, and $p_{\text{hit}}(i)$ is the risk for the i -th inspection interval. This penalty is also incorporated into the penalty-augmented objective function.

3. Monte Carlo simulation architecture

The Monte Carlo simulation framework used in this study is designed to evaluate and compare the performance of three adaptive condition-based maintenance (CBM) strategies. The framework simulates the system's degradation and maintenance processes under varying operational conditions, enabling the evaluation of each strategy's impact on system reliability, cost, downtime, and maintenance actions. The framework incorporates a stochastic degradation model based on a Wiener process, and Monte Carlo simulations are employed to account for the randomness inherent in the system's degradation and the effects of maintenance strategies [56] [57] [30].

Each simulation run models the system's degradation from an initial "new" state, $X(0)=0$, up to a predefined operational horizon of 20000 time units. The degradation is simulated at discrete time steps using the Wiener process, governed by the stochastic differential equation (SDE) presented in Equation (1), where $X(t)$ represents the degradation state of the system at time t , μ is the drift coefficient indicating the average degradation rate, σ is the diffusion coefficient representing the uncertainty or variability of degradation, and $dW(t)$ is the increment of a standard Wiener process, introducing randomness into the degradation trajectory [58] [59].

The system is considered to have failed when its degradation state exceeds a predefined failure threshold L_f , expressed by the condition presented in Equation (2).

Upon failure, corrective maintenance (CM) is triggered, and the system is reset to its initial state. The system's degradation process is tracked continuously until failure occurs or until the operational horizon is reached.

Three maintenance strategies are evaluated: the fixed-interval policy ($\Delta T, M$), the adaptive-interval policy ($\Delta T_k, M$), and the adaptive-threshold policy ($\Delta T, M_k$). These strategies determine when preventive maintenance (PM) is performed and how inspection intervals or thresholds are adjusted based on the system's degradation state.

The fixed-interval policy schedules inspections at fixed intervals ΔT ,

regardless of the system's degradation. Preventive maintenance is triggered when the degradation state exceeds a fixed threshold L_m . This strategy maintains constant inspection intervals, independent of the system's condition.

In the adaptive-interval policy, the inspection interval ΔT_k is adjusted based on the system's remaining useful life (RUL). The RUL at time t is computed in Equation (4).

Where μ is the drift coefficient. The inspection interval ΔT_k is then adjusted according to the formula of Equation (5), this policy dynamically shortens the inspection interval as the system approaches failure and lengthens it when the system is in a better condition.

The adaptive-threshold policy adjusts the threshold for preventive maintenance $L_{m,k}$ based on the system's degradation state. The threshold is recalculated at each inspection epoch according to Equation (6), where α_L is the adjustment parameter, $L_{\min}$ is the minimum allowable threshold, and δL_{safety} ensures that the threshold remains sufficiently below the failure limit, maintaining a safety margin. Preventive maintenance is performed when the degradation state exceeds the dynamically adjusted threshold $L_{m,k}$.

The cost model incorporates the costs of preventive maintenance, corrective maintenance, and inspections, as well as the downtime associated with each action. The total cost for a simulation run is computed by summing the costs of all preventive and corrective maintenance actions, and inspections, along with the associated downtime. Each maintenance action incurs specific costs: C_{PM} for preventive maintenance, C_{CM} for corrective maintenance, and C_{insp} for inspections.

Reliability is evaluated using the Kaplan–Meier survival estimate. The survival function $S(t)$ represents the probability that the system remains operational up to a specified time t , and reliability is determined by Equation (7).

Where $S(t)$ is the survival probability. The reliability requirement is set to $R_{\min}=0.95$, and each strategy is assessed based on whether it meets or exceeds this reliability threshold at the reporting horizon.

The between-inspection risk p_{hit} is the probability that the system's degradation exceeds the failure threshold L_f before the next inspection. This risk is calculated using Equation (8). The goal of the maintenance strategies is to minimize this risk, ensuring that the system does not fail between inspections.

Penalties are introduced for reliability shortfalls and excessive between-inspection risk. The penalty for reliability shortfall is calculated if the system's reliability at the reporting horizon t_{report} falls below the required threshold $R_{\min}$. The penalty is given by Equation (9). This penalty is incorporated into the overall evaluation of each strategy's effectiveness.

The penalty for excessive between-inspection risk is calculated for each inspection interval as the probability that the system's degradation exceeds the failure threshold between inspections. The penalty for each inspection interval is given by Equation (10). This penalty is used to evaluate how well the strategy minimizes the risk of failure occurring between inspections.

Each simulation run is replicated 1000 times to ensure statistical reliability. For each replication, the performance metrics, such as total cost, downtime, reliability, and maintenance actions, are calculated. These metrics are then averaged across all replications to provide robust estimates of each strategy's performance.

To ensure reproducibility, all simulations are performed using a fixed random seed. Independent random number generators are used for each maintenance strategy to facilitate unbiased comparisons and ensure that the differences in performance are due to the strategies themselves rather than random variation.

4. Simulation configuration and input parameters

This section outlines the configuration of the Monte Carlo simulations used to assess the performance of three maintenance strategies: the fixed-interval policy ($\Delta T, M$), the adaptive-interval policy ($\Delta T_k, M$), and

the adaptive-threshold policy $(\Delta T, M_k)$. The simulation framework is based on a stochastic degradation model with a Wiener process, and adaptive maintenance strategies are incorporated to evaluate their impacts on system reliability, cost, and downtime.

The simulations involved 1000 independent replications, which ensures statistical robustness. Each simulation models the degradation process of the system over a total operational horizon of 20000 time units, with an integration step size of 1 time unit. The degradation of the system is modeled as a Wiener process, with the degradation state $X(t)$ at time t governed by the stochastic differential equation (1).

Where the system is considered to have failed when $X(t)$ exceeds the failure threshold $L_f=5$, at which point corrective maintenance is triggered. The initial state $X(0)=0$ represents a “new” system with no degradation.

For the fixed-interval policy $(\Delta T, M)$, inspections are scheduled at a fixed interval of 120 time units, independent of the system's degradation state. Preventive maintenance (PM) is performed when the degradation level $X(t)$ exceeds a fixed threshold of $L_m=3.5$. This policy does not consider the system's current degradation rate and relies on a regular inspection schedule, which may result in either unnecessary preventive actions or missed failures.

In contrast, the adaptive-interval policy $(\Delta T_k, M)$ adjusts the inspection intervals based on the system's degradation state. The interval between inspections is determined by the system's remaining useful life (RUL), calculated as the time remaining until the failure threshold is reached. The inspection interval ΔT_k is given by Equation (5), where $\alpha_T=0.4$ controls the degree of adaptation, and $\Delta T_{min}=20$ and $\Delta T_{max}=360$ represent the minimum and maximum allowable intervals, respectively. This adaptive strategy aims to shorten inspection intervals as the system nears failure and to lengthen intervals when the system is in better condition, which can improve cost-efficiency while maintaining system reliability.

The adaptive-threshold policy $(\Delta T, M_k)$ adjusts the threshold for preventive maintenance dynamically based on the system's RUL. The threshold $L_{m,k}$ is recalculated at each inspection epoch using Equation (6), where $\alpha_L=0.3$ is the adaptation parameter, $L_{min}=2.5$ is the lower bound for the threshold, and $\delta L_{safety}=0.4$ ensures that the threshold remains sufficiently far from the failure limit. This strategy aims to adjust the preventive maintenance threshold earlier when the system is in a deteriorated state, allowing for proactive maintenance actions that may prevent unplanned corrective maintenance.

The cost structure is defined for preventive maintenance, corrective maintenance, and inspections. Each preventive maintenance action incurs a cost of 100 units, while each corrective maintenance action costs 5000 units, which includes the repair cost and associated penalties. Preventive maintenance results in a downtime of 30 time units, while corrective maintenance results in 800 time units of downtime. Each inspection incurs a cost of 5 units, with a downtime of 5 time units.

Reliability is assessed by computing the Kaplan–Meier survival estimate at a reporting horizon of 2000 time units, denoted as $S(t)$. The reliability requirement is set at $R_{min}=0.95$, and each strategy is evaluated based on its ability to meet or exceed this threshold. The between-inspection risk, p_{hit} , is defined as the probability that the system's degradation state exceeds the failure threshold L_f between two consecutive inspections. This measure is used to evaluate the effectiveness of the strategies in mitigating the risk of failure between inspections.

To ensure the reproducibility of the simulation results, a fixed random seed value is set at the beginning of the simulation process. In this study, we use a seed value of 424242. This ensures that every time the simulation is run with this seed, the random number generation follows the same sequence, thereby allowing for a consistent and fair comparison of the strategies under evaluation. By fixing the seed value to 424242, we eliminate variability that could arise from different random sequences, allowing for an unbiased comparison of the adaptive maintenance strategies across all replications. This approach is essential for obtaining reliable and consistent results in Monte Carlo simulations.

A total of 1000 replications were run for each strategy, and the performance of each strategy was evaluated based on the mean values of key metrics across these replications.

The simulation parameters used in this study are selected to ensure physical consistency and to reflect typical operating conditions reported in the condition-based maintenance literature. Rather than being derived from a specific dataset or reference, these values are chosen within realistic ranges to represent the relative behavior of maintenance processes.

In particular, the relationship between preventive and corrective maintenance costs is defined such that corrective actions are significantly more expensive and involve longer downtime, which is a standard assumption in maintenance modeling. Similarly, the inspection intervals and adaptive parameters are selected to allow meaningful variation in inspection frequency and maintenance decisions, while remaining within feasible operational limits.

The purpose of this study is to compare the structural performance of different maintenance strategies under identical conditions. Therefore, all policies are evaluated using the same parameter set, ensuring a fair and consistent comparison. The conclusions drawn are thus related to the inherent characteristics of the strategies rather than to specific parameter tuning.

The adaptive maintenance strategies depend on several parameters, including α_T and α_L , which control the intensity of interval and threshold adaptation, as well as constraint bounds ΔT_{min} , ΔT_{max} , and L_{min} . In this study, these parameters are selected within physically meaningful ranges and applied through bounded update rules, ensuring that inspection intervals and maintenance thresholds remain within feasible limits.

Instead of performing a formal parameter optimization or grid-search, the robustness of the proposed strategies is evaluated using Monte Carlo simulation. By simulating a large number of stochastic degradation trajectories, the approach captures variability in system dynamics and provides a statistical assessment of performance under uncertainty. The results indicate that the relative behavior of the strategies remains stable across simulations, suggesting that the conclusions are not sensitive to specific parameter choices.

This robustness is further confirmed by the consistency of performance indicators observed across all simulation results presented in Section 5. A systematic sensitivity analysis or optimization framework could further refine parameter selection and is identified as an important direction for future research.

To assess the statistical reliability of the Monte Carlo estimates, confidence intervals were computed for the main performance metrics, including the average cost rate and the mean between-inspection risk. For each metric, a 95% confidence interval was constructed using the sample mean and standard deviation obtained from the 1000 independent simulation runs, assuming approximate normality based on the central limit theorem.

The resulting confidence intervals were found to be relatively narrow, indicating good convergence of the estimates and limited sampling variability. This confirms that the number of Monte Carlo replications used in this study is sufficient to provide stable and reliable performance evaluations.

While more advanced techniques such as bootstrap resampling or variance-reduction methods could further refine the statistical analysis, the present approach provides a consistent and computationally efficient assessment of uncertainty.

5. Results and interpretations

This section presents and interprets the simulation outcomes obtained for the three maintenance decision policies, namely $(\Delta T, M)$, $(\Delta T_k, M)$, and $(\Delta T, M_k)$. The analysis integrates statistical evidence from Tables 1–6 with graphical representations in Figs. 4a–15c, and emphasizes comparative performance across the strategies in terms of cost,

Table 1

Global summary.

Strategy	C_{avg} (long-run average cost rate)	s_C (std. dev. of cost rate)	$\rho_{PM/CM}$ (ratio of PM to CM counts)	D_{avg} (total downtime per run)	I_{avg} (inspections per run)	R (Reliability by Kaplan–Meier survival at R_{time})	$1[R \geq R_{min}]$ (meets required reliability)
$(\Delta T, M)$	4.4868	0.87694	1.5832	15388	154.27	0.165	false
$(\Delta T_k, M)$	4.4526	0.8733	1.6076	15282	154.38	0.161	false
$(\Delta T, M_k)$	4.4648	0.84224	1.5876	15317	154.34	0.157	false

Table 2

Mean PM and CM.

Strategy	E[PM] (expected preventive actions per run)	E[CM] (expected corrective actions per run)
$(\Delta T, M)$	27.305	17.247
$(\Delta T_k, M)$	27.5	17.106
$(\Delta T, M_k)$	27.243	17.16

Table 3

PM/CM ratio.

Strategy	$\rho_{PM/CM}$ (ratio of mean PM to CM counts)
$(\Delta T, M)$	1.5832
$(\Delta T_k, M)$	1.6076
$(\Delta T, M_k)$	1.5876

Table 4Short-horizon risk p_{hit} .

Strategy	E[p_{hit}] (mean hitting probability)	$P_{50(p_{hit})}$ (median of p_{hit})	$P_{90(p_{hit})}$ (90th percentile of p_{hit})	$P_{99(p_{hit})}$ (99th percentile of p_{hit})	$Pr[p_{hit} > \beta]$ (fraction exceeding $\beta = 0.05$)
$(\Delta T, M)$	0.2539	0.12779	0.73426	0.95776	0.64086
$(\Delta T_k, M)$	0.25584	0.13026	0.73807	0.95659	0.64476
$(\Delta T, M_k)$	0.2543	0.12946	0.73372	0.9586	0.64282

Table 5

Trigger statistics and safety margins.

Strategy	E[$L_{m,k}$] (mean adaptive PM trigger level)	$L_{m(P10)}$ (10th percentile of $L_{m,k}$)	$L_{m(P90)}$ (90th percentile of $L_{m,k}$)	E[$L_f - L_{m,k}$] (mean safety margin)	Margin $_{P10}$ (10th percentile safety margin)
$(\Delta T, M)$	2.7333	2.5	3.1867	2.2667	1.8133
$(\Delta T_k, M)$	2.7352	2.5	3.19	2.2648	1.81
$(\Delta T, M_k)$	2.7338	2.5	3.1862	2.2662	1.8138

reliability, downtime, inspection burden, and risk exposure.

5.1. Degradation trajectories

The temporal evolution of the degradation process under the three maintenance policies is reported in Fig. 4a–c. Each panel presents a subset of representative trajectories obtained from the Monte Carlo

Table 6Penalty-augmented objective J_λ .

Strategy	C_{avg} (average cost component in J_λ)	Var[C] (variance of cost)	R_{time} (survival at $R_{time} = 2000$)	$R_{min} - R$ (reliability gap with $R_{min} = 0.95$)	Penalty $_R$ ($\lambda_R \cdot \text{shortfall}$, $\lambda_R = 10000$)	$Pr[p_{hit} > \beta]$ (Fraction of p_{hit} of beta with $\beta = 0.05$)	Penalty $_{hit}$ ($\lambda_{hit} \cdot Pr$, $\lambda_{hit} = 1000$)	J_{lambda} (J_λ , penalty-augmented objective)
$(\Delta T, M)$	4.4868	0.76826	0.165	0.785	7850	0.64086	640.86	8495.4
$(\Delta T_k, M)$	4.4526	0.7619	0.161	0.789	7890	0.64476	644.76	8539.2
$(\Delta T, M_k)$	4.4648	0.70867	0.157	0.793	7930	0.64282	642.82	8577.3

simulations, with the preventive maintenance (PM) threshold and the hard failure boundary superimposed to facilitate interpretation of intervention dynamics.

Under the fixed-interval baseline policy $(\Delta T, M)$ (Fig. 4a), inspections are scheduled uniformly with period $\Delta T = 120$. The inspection cadence is independent of the stochastic variability of the degradation process. This structural rigidity produces two recurring patterns. When the degradation rate is unusually high, the constant cadence permits the process to cross the failure boundary L_f before the next scheduled inspection, thereby generating corrective maintenance (CM). When the degradation rate is unusually low, inspections are performed more frequently than required, increasing operational cost without contributing significantly to reliability. These features are consistent with the moderate PM/CM ratio and the relatively high cost variance observed later in Tables 2 and 3.

The adaptive-interval policy $(\Delta T_k, M)$ (Fig. 4b) introduces state dependence into the inspection schedule. At each inspection epoch, the subsequent interval is adjusted as a function of the estimated remaining useful life (RUL). This produces denser inspections when trajectories approach the critical boundary and more relaxed scheduling when degradation evolves slowly. The effect is visible in the clustering of inspection epochs around steep degradation segments. This mechanism reduces the probability of missed failures and increases the likelihood of preventive interventions, a trend confirmed quantitatively in Tables 2 and 3, where the strategy attains the lowest average number of CM events and the highest PM/CM ratio.

The adaptive-threshold policy $(\Delta T, M_k)$ (Fig. 4c) maintains a fixed inspection cadence while adjusting the intervention threshold $L_{m,k}$ as a function of the estimated RUL. This produces earlier interventions in certain trajectories, particularly those with accelerated degradation. However, the adaptive threshold is frequently constrained by feasibility bounds ($L_{min} = 2.5$, $L_f - \delta L_{safety} = 4.6$), as confirmed in Table 5. The saturation of the adaptive rule limits the effective flexibility of the policy, which explains why overall performance indicators, including mean cost and PM/CM ratio, remain close to those of the fixed-threshold baseline.

The visual evidence provided in Fig. 4a–c indicates that interval adaptation modifies the inspection process in a manner that more closely reflects the stochastic dynamics of degradation, whereas threshold adaptation exerts only a limited effect due to structural constraints. The distinct patterns observed at this stage provide a basis for understanding the differences in cost, downtime, reliability, and risk indicators discussed in the following subsections.

5.2. Cost and downtime distributions

The distributions of the average cost rate are presented in Fig. 5a–c, and the corresponding distributions of total downtime are shown in

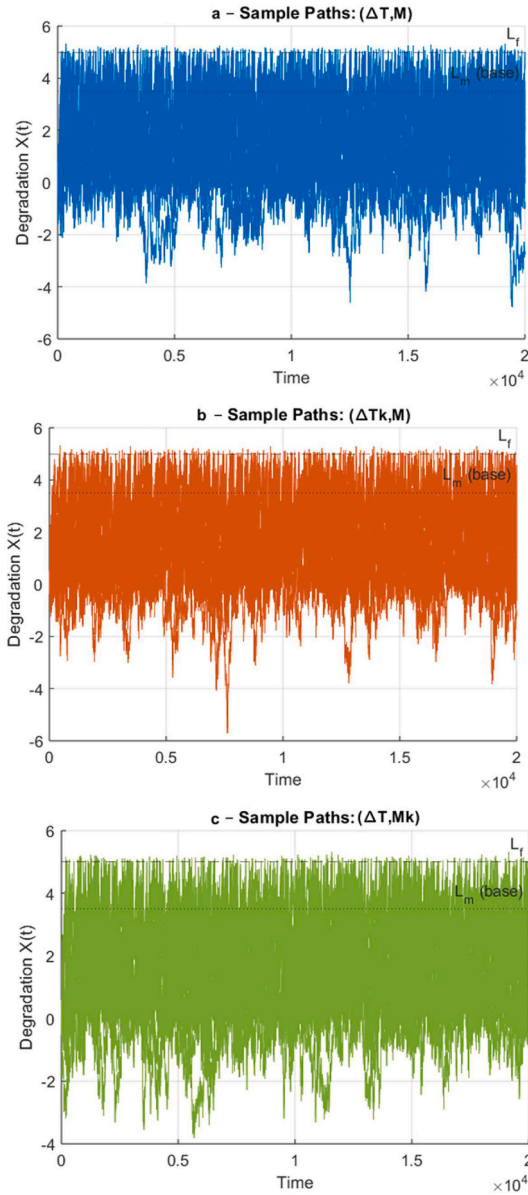


Fig. 4. Sample paths for all strategies.

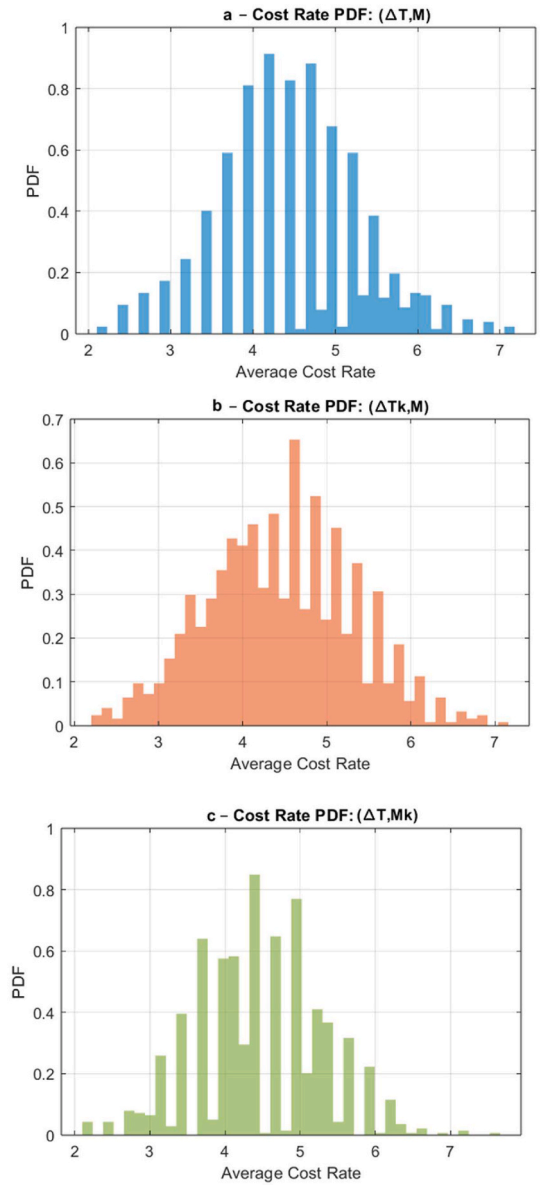


Fig. 5. Cost distributions for all strategies.

Fig. 6a–c. Numerical results are reported in Table 1. The three strategies exhibit comparable outcomes in terms of both cost and downtime; however, consistent differences are observed when the results are compared across panels.

For the fixed-interval policy $(\Delta T, M)$ (Figs. 5a and 6a), the mean cost rate equals 4.4868 with a standard deviation of 0.8769. The average downtime is 15388, and the mean number of inspections per run is 154.27. Reliability at the reporting horizon of 2000 time units is 0.165, which is far below the required threshold of 0.95. The distributions exhibit wide dispersion, with extended tails towards high values. These features reflect the rigidity of the fixed inspection schedule: rapid degradation trajectories often result in late detection and corrective maintenance, whereas slower trajectories incur unnecessary inspection costs without corresponding improvements in reliability.

The adaptive-interval policy $(\Delta T_k, M)$ (Figs. 5b and 6b) achieves improved performance. The mean cost decreases to 4.4526 with a standard deviation of 0.8733, and the average downtime reduces to 15282. The number of inspections (154.38) remains comparable to that of the baseline, which indicates that improvements are achieved not through reduced inspection frequency but through more effective

temporal allocation of inspections. Reliability at the reporting horizon is 0.161, which is statistically indistinguishable from the baseline. The histograms in Figs. 5b and 6b are shifted slightly to the left relative to those in Figs. 5a and 6a, with higher density in the low-cost and low-downtime regions. This shift is consistent with the higher *PM/CM* ratio of 1.6076 reported in Table 1, which confirms that adaptive interval scheduling increases the likelihood of preventive interventions and reduces the frequency of costly corrective events.

The adaptive-threshold policy $(\Delta T, M_k)$ (Figs. 5c and 6c) provides results that are only marginally better than the baseline. The mean cost rate is 4.4648 with a standard deviation of 0.8422, and the mean downtime equals 15317. The mean number of inspections is 154.34, very close to the other strategies. Reliability at the reporting horizon is 0.157, the lowest among the three policies. The shapes of the distributions in Figs. 5c and 6c are nearly identical to those in Figs. 5a and 6a, with only a minor leftward displacement. This outcome corresponds to the evidence in Table 5, which shows that the adaptive threshold is frequently constrained by feasibility bounds, limiting its operational effect on cost and downtime.

The comparative results in Fig. 5a–c and 6a–6c, reinforced by the

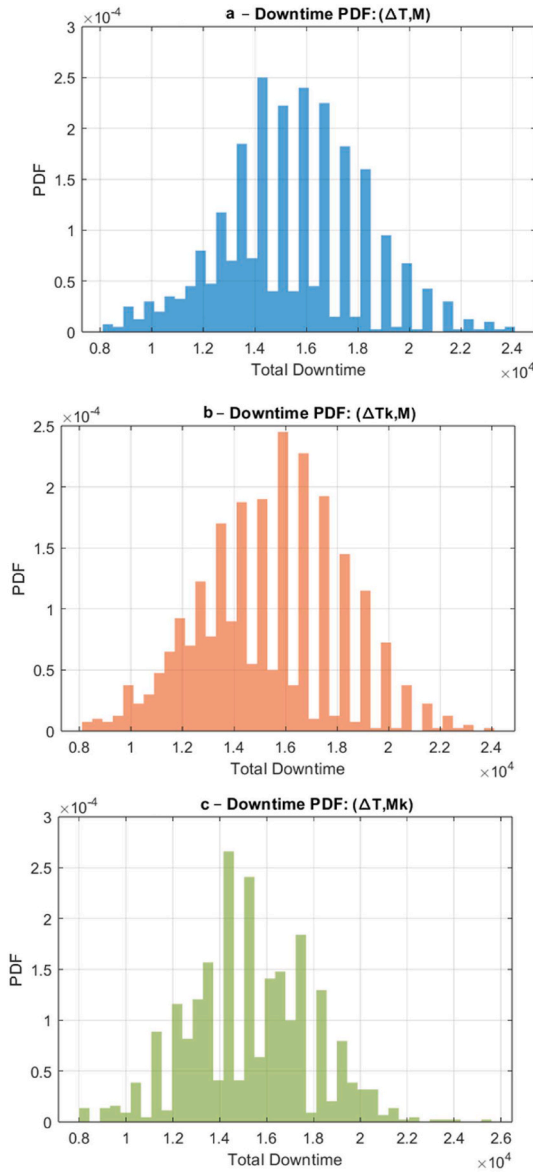


Fig. 6. Downtime distributions for all strategies.

numerical values in Table 1, establish a consistent ordering. The adaptive-interval policy $(\Delta T_k, M)$ yields the most favorable cost and downtime outcomes. The adaptive-threshold policy $(\Delta T, M_k)$ performs slightly better than the baseline but remains close to it. The fixed-interval baseline $(\Delta T, M)$ is associated with the least advantageous results. Despite these differences, all three strategies exhibit very low reliability at the reporting horizon, demonstrating that improvements in adaptivity, under the present parameterization, do not suffice to ensure compliance with reliability requirements.

5.3. Preventive and corrective actions

The distributions of inspection counts, preventive maintenance (PM), and corrective maintenance (CM) events are shown in Fig. 7a–c. Numerical averages are reported in Tables 2 and 3, providing complementary quantitative evidence.

For the fixed-interval baseline policy $(\Delta T, M)$ (Fig. 7a), the average number of inspections is 154.27 per run. The mean PM count is 27.305, while the mean CM count reaches 17.247 (Table 2). The resulting PM/CM ratio is 1.5832 (Table 3). The boxplots exhibit considerable dispersion, reflecting variability in degradation trajectories. The

relatively high CM frequency indicates that the fixed inspection cadence often fails to anticipate accelerated degradation, which leads to unplanned corrective interventions despite the presence of scheduled preventive activities.

The adaptive-interval policy $(\Delta T_k, M)$ (Fig. 7b) modifies the distribution of actions in a favorable direction. The mean number of inspections is 154.38, essentially identical to the baseline, which shows that improvements are not obtained by reducing inspection effort but rather by altering its temporal allocation. The mean PM count increases slightly to 27.5, while the mean CM count decreases to 17.106 (Table 2). The PM/CM ratio consequently improves to 1.6076 (Table 3). The boxplots confirm a downward shift in CM occurrences relative to the baseline. This evidence demonstrates that interval adaptation enhances the probability of preventive interventions in high-risk states, thereby reducing the reliance on corrective maintenance.

The adaptive-threshold policy $(\Delta T, M_k)$ (Fig. 7c) produces outcomes very similar to the baseline. The mean PM count is 27.243 and the mean CM count is 17.160, yielding a PM/CM ratio of 1.5876 (Tables 2 and 3). The distributions of inspections, PM, and CM in Fig. 7c are nearly indistinguishable from those in Fig. 7a. This lack of differentiation is consistent with the behavior described in Section 5.1 and in Table 5, where the adaptive threshold often saturates at feasibility bounds, thereby limiting its effect on intervention timing.

$(\Delta T_k, M) > (\Delta T, M_k) \approx (\Delta T, M)$ Comparison of the three strategies indicates that interval adaptation $(\Delta T_k, M)$ produces measurable gains in the balance between preventive and corrective maintenance, while threshold adaptation $(\Delta T, M_k)$ offers only marginal changes relative to the baseline. The ranking established by Tables 2 and 3 is clear:

The analysis confirms that adaptive adjustment of inspection intervals is a more effective mechanism than threshold adjustment alone for influencing the distribution of maintenance actions under the present conditions.

5.4. Reliability assessment at reporting horizon

The Kaplan–Meier survival estimates are reported in Fig. 8, and the reliability values at the reporting horizon of 2000 time units are summarized in Table 1. The results demonstrate a rapid decline in survival probability under all three policies, with no strategy satisfying the required threshold of $R_{\min}=0.95$.

The Kaplan–Meier curves shown in Fig. 8 provide a time-dependent representation of system reliability across the full operational horizon. The survival functions for the three strategies exhibit very similar decay patterns, with a rapid decline in reliability at early stages. This indicates that failures occur predominantly in the initial phase of operation, regardless of the maintenance policy applied.

The close overlap of the survival curves suggests that differences between the strategies are not statistically significant in terms of time-dependent reliability behavior. This observation is consistent with the similar reliability values reported at the reporting horizon. Although formal statistical tests such as the log-rank test or hazard rate analysis could be applied to compare survival distributions, the visual and numerical evidence indicates that such differences would be minimal.

These results confirm that the primary limitation of all strategies lies in their inability to control early failures driven by between-inspection risk, rather than in differences in long-term degradation behavior.

For the fixed-interval policy $(\Delta T, M)$, the estimated survival probability at $t=2000$ is 0.165. The survival curve declines steeply in the early portion of the horizon, reflecting the frequent occurrence of failures between scheduled inspections. The fixed inspection cadence does not adequately respond to trajectories characterized by accelerated degradation, which explains the relatively high proportion of corrective maintenance events observed in previous subsections.

The adaptive-interval policy $(\Delta T_k, M)$ yields a survival probability of 0.161 at the reporting horizon. The survival profile in Fig. 8 closely overlaps with that of the baseline. The inability of interval adaptation to

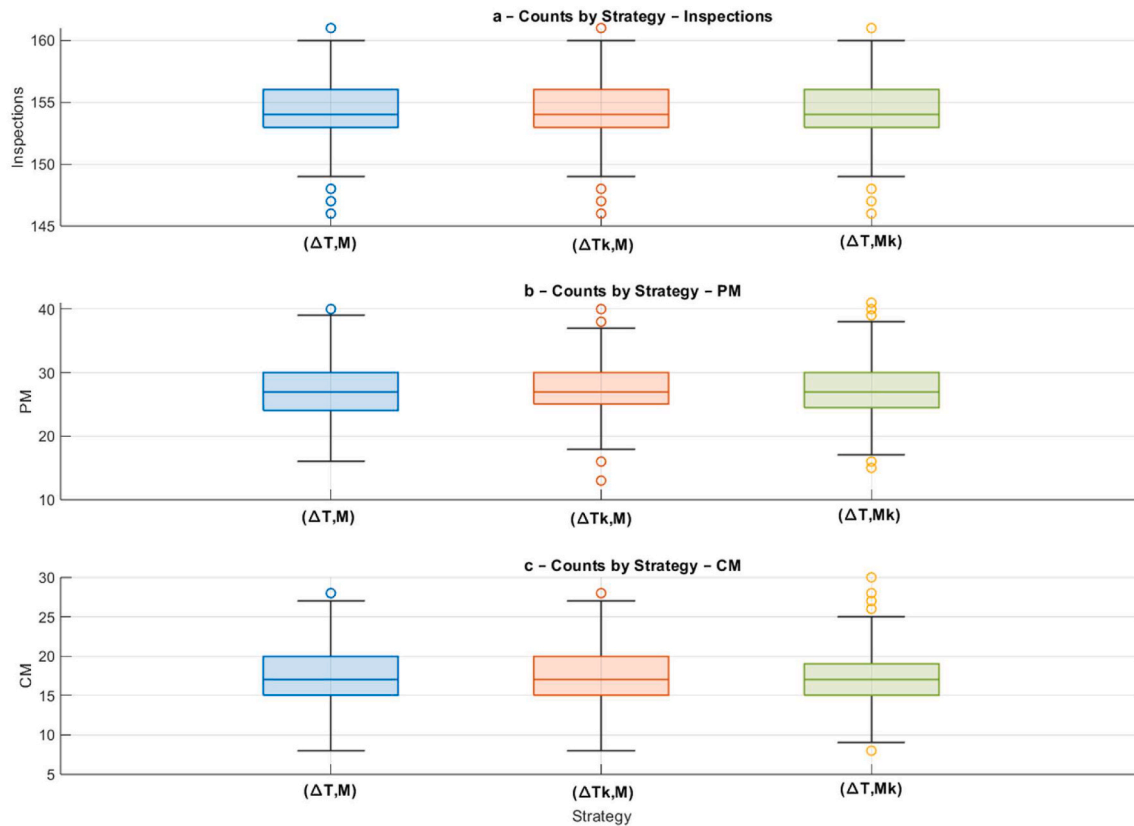


Fig. 7. Maintenance action counts for all strategies.

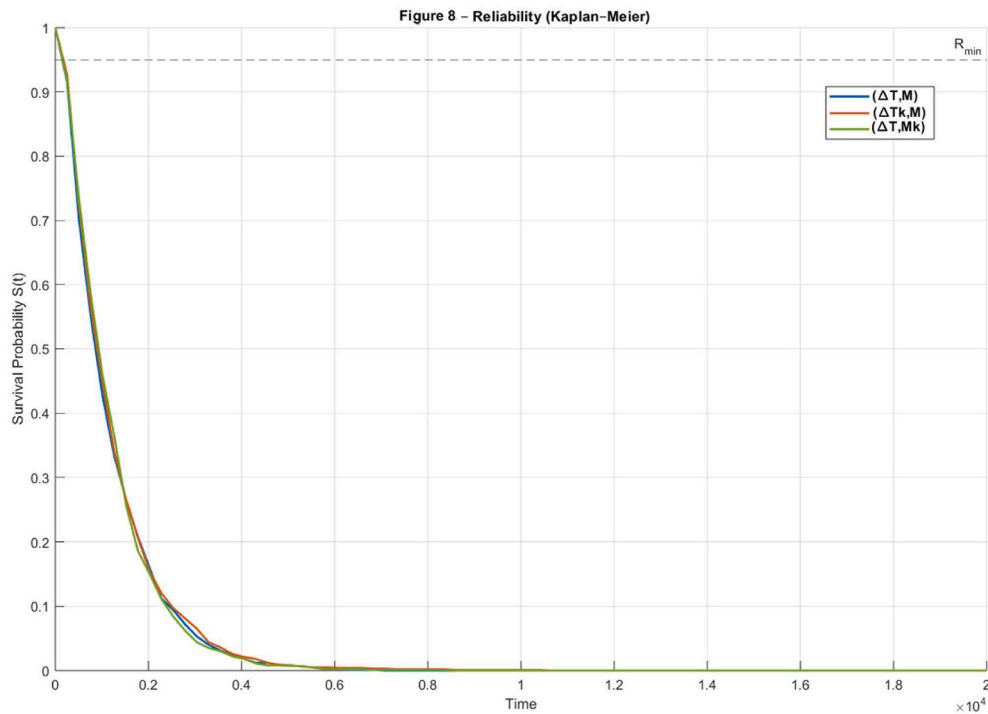


Fig. 8. Reliability (Kaplan-Meier) for all strategies.

improve reliability is consistent with the observation that, although inspection intervals are adjusted, the magnitude of adaptation is insufficient to prevent early failures. The improvements previously observed in cost and downtime are therefore achieved without a corresponding

gain in reliability.

The adaptive-threshold policy $(\Delta T, M_k)$ records the lowest reliability among the three strategies, with a survival probability of 0.157 at $t=2000$. The survival curve again follows the same declining trend as

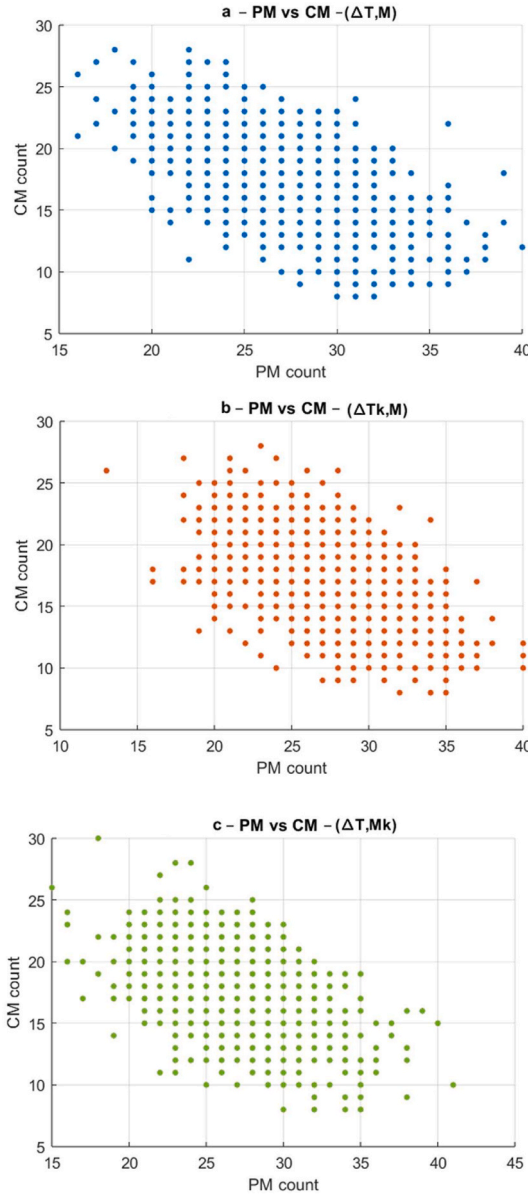


Fig. 9. PM-CM scatter for all strategies.

the other two strategies. This outcome reflects the operational constraints on the adaptive threshold, which frequently saturates at imposed bounds and does not provide sufficient flexibility to anticipate failure.

The comparison of the three strategies indicates that reliability performance remains poor regardless of the policy applied. The differences among the reported values (0.165, 0.161, and 0.157) are small in magnitude and do not alter the fundamental conclusion that the reliability requirement is not achieved under any of the evaluated configurations.

The fact that none of the evaluated strategies satisfies the reliability requirement ($R_{\min} = 0.95$) can be explained by the inherent limitations of inspection-based maintenance under stochastic degradation. In the proposed framework, failures may occur between inspection epochs due to the random evolution of the degradation process, resulting in significant between-inspection risk.

Although adaptive interval strategies adjust inspection frequency based on the system state, this mechanism primarily redistributes inspection efforts rather than eliminating high-risk intervals. As a result, early failures remain frequent, leading to a rapid decline in survival

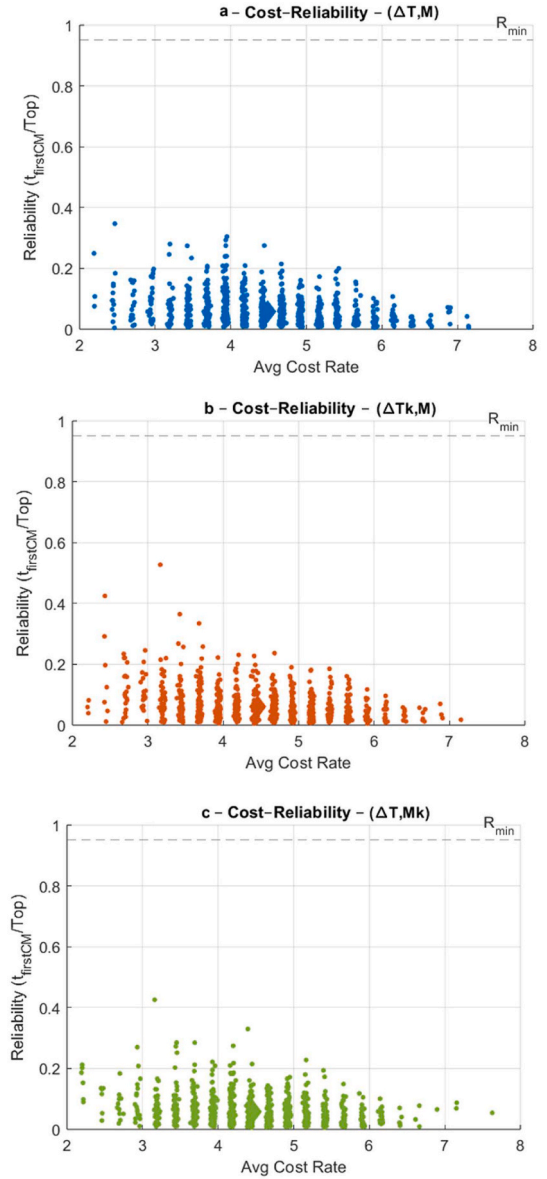


Fig. 10. Cost-reliability scatter for all strategies.

probability, as observed in Fig. 8.

The reliability target considered in this study represents a stringent requirement typical of safety-critical systems. The inability of the evaluated strategies to meet this target does not indicate a flaw in the model, but rather highlights a structural limitation of inspection-based maintenance approaches. Achieving higher reliability levels would likely require more advanced strategies, such as continuous condition monitoring, predictive models, or real-time decision frameworks capable of reducing between-inspection failure risk.

5.5. Preventive maintenance (PM) and corrective maintenance (CM)

The relationship between preventive maintenance (PM) and corrective maintenance (CM) is illustrated in Fig. 9a-c. Each subplot represents the joint distribution of PM and CM events for the three strategies, with simulation outcomes shown as scatter points.

For the fixed-interval policy ($\Delta T, M$) (Fig. 9a), the scatter distribution is widely dispersed, with average values of 27.305 p.m. and 17.247 CM events per run (Table 2). The points form a diffuse diagonal trend, indicating that higher PM counts are associated with modest reductions

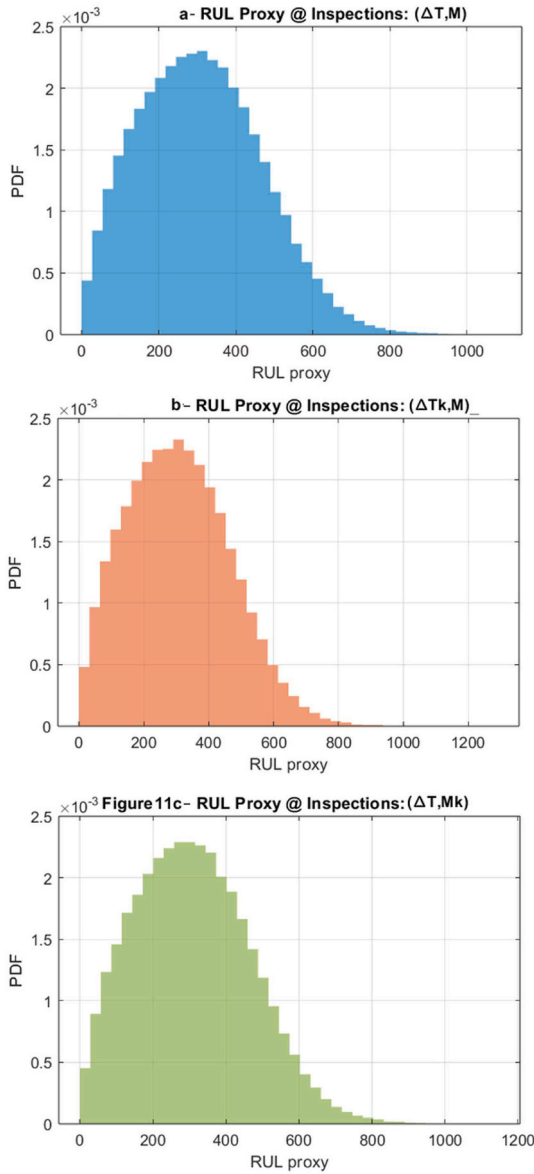


Fig. 11. RUL proxy distributions for all strategies.

in CM counts. The absence of a strong negative correlation highlights the structural limitation of a fixed inspection cadence: preventive actions occur frequently, yet corrective events remain substantial.

The adaptive-interval policy $(\Delta T_k, M)$ (Fig. 9b) shifts the scatter distribution upward and leftward relative to the baseline. The mean PM count increases to 27.5, while the mean CM count decreases to 17.106 (Table 2). This adjustment yields the highest PM/CM ratio among the strategies (1.6076; Table 3). The point cloud exhibits greater density in regions with relatively high PM and low CM, reflecting the ability of shortened inspection intervals near critical states to facilitate preventive interventions that avert failures.

The adaptive-threshold policy $(\Delta T, M_k)$ (Fig. 9c) produces a distribution that closely resembles the baseline. The averages, 27.243 p.m. and 17.16 CM (Table 2), are nearly identical to those of the fixed-interval policy. The scatter pattern does not deviate significantly from that in Fig. 9a, and the PM/CM ratio of 1.5876 (Table 3) confirms the limited impact of threshold adjustment.

The comparison across the three panels shows that interval adaptation modifies the balance between preventive and corrective actions, while threshold adaptation does not. The results indicate that $(\Delta T_k, M)$

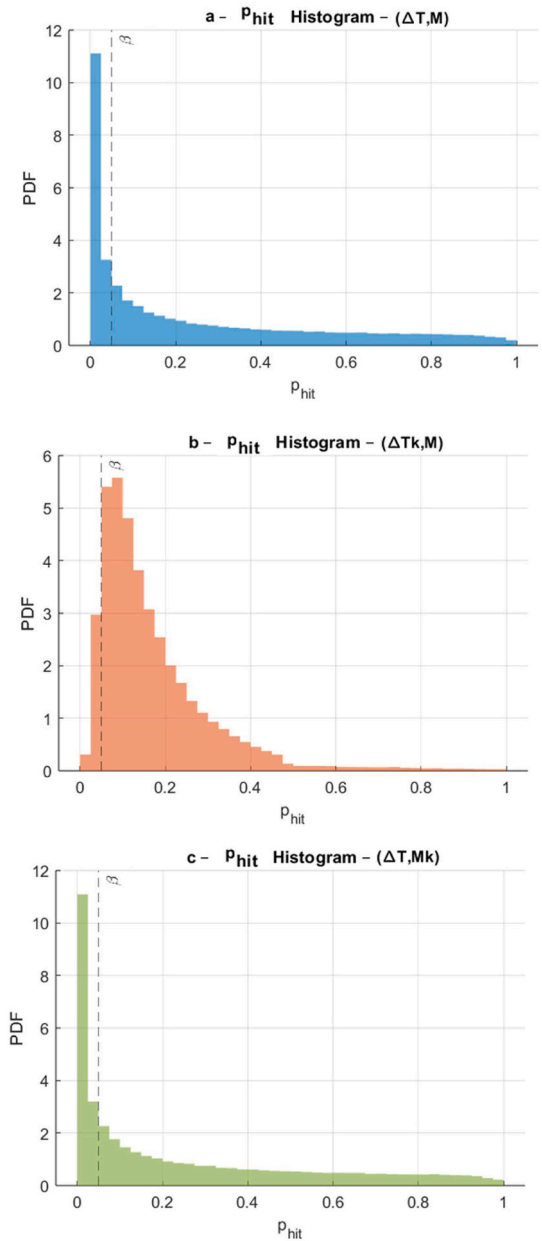


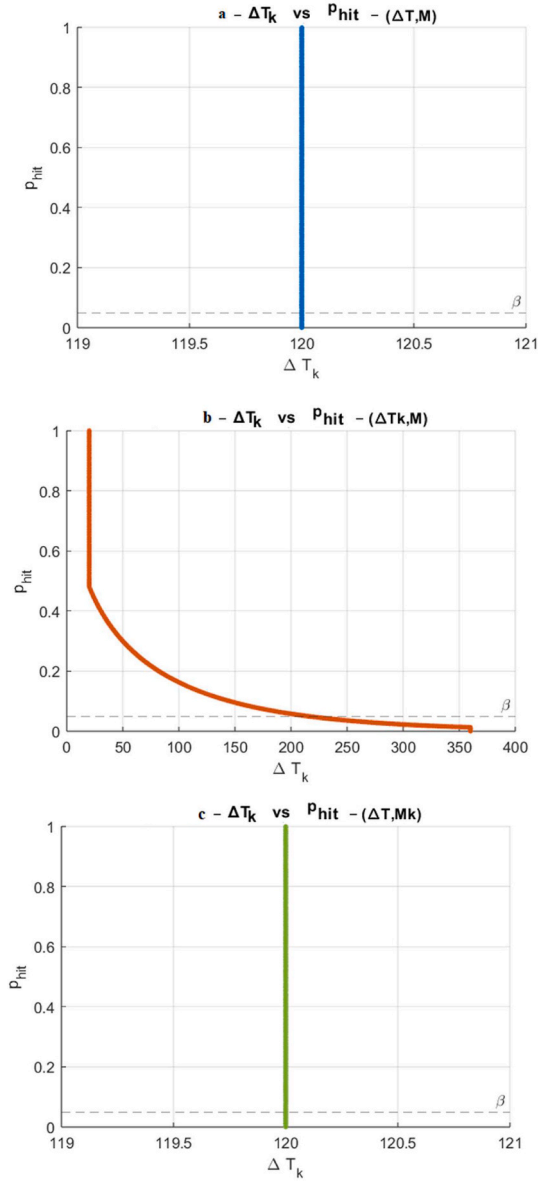
Fig. 12. P_{hit} Histogram for all strategies.

achieves a more favorable substitution of preventive for corrective maintenance, whereas $(\Delta T, M)$ and $(\Delta T, M_k)$ yield almost indistinguishable outcomes.

5.6. Cost-reliability trade-offs

The interaction between average cost rate and reliability performance is illustrated in Fig. 10a–c, where each point corresponds to the outcome of a single simulation run. Reliability is expressed as the ratio of the time to first corrective maintenance (CM) event to the total operational horizon. The mean values of cost and reliability for each strategy are indicated by diamond markers, and the numerical values are reported in Table 1.

For the fixed-interval policy $(\Delta T, M)$ (Fig. 10a), the mean cost rate is 4.4868 and the mean reliability proxy is 0.165. The scatter points are broadly dispersed, with a clear tendency for low-cost realizations to be associated with shorter times to failure. Runs with higher preventive activity generate higher cost but also extend the time to first CM. This

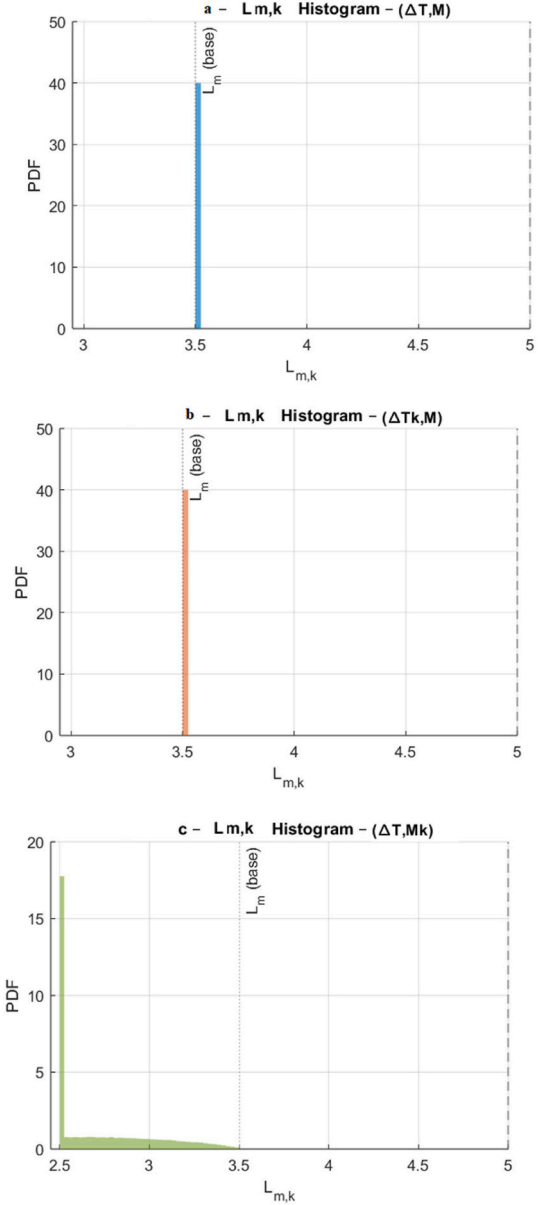
Fig. 13. ΔT_k vs P_{hit} for all strategies.

dispersion highlights the inherent trade-off of fixed inspection scheduling: lower maintenance expenditure comes at the expense of reliability, while improved reliability requires additional preventive interventions.

The adaptive-interval policy ($\Delta T_k, M$) (Fig. 10b) shows a distribution shifted slightly toward the lower-cost region compared with the baseline. The mean cost decreases to 4.4526, while the mean reliability proxy remains at 0.161. The pattern of dispersion remains similar to that of the baseline, but the concentration of points in the moderate-to-low cost region is more pronounced. This outcome indicates that interval adaptation reduces the frequency of costly corrective events without generating measurable improvements in reliability at the system level.

The adaptive-threshold policy ($\Delta T, M_k$) (Fig. 10c) produces results that are nearly indistinguishable from those of the baseline. The mean cost rate is 4.4648, and the mean reliability proxy is 0.157. The scatter distribution maintains the same spread and orientation as in Fig. 10a, with no visible reduction in the association between low cost and early failures. The slight cost improvement relative to the baseline does not translate into a material gain in reliability.

The comparison across the three strategies indicates that interval

Fig. 14. $L_{m,k}$ Histogram for all strategies.

adaptation achieves a marginal reduction in cost without altering the reliability profile, whereas threshold adaptation has little effect in either dimension. The fixed-interval baseline remains more costly than the adaptive-interval policy and slightly more reliable than the adaptive-threshold policy. None of the strategies achieve the reliability requirement of $R_{min}=0.95$, which demonstrates the persistence of early failures driven by between-inspection risk.

5.7. Remaining useful life (RUL) proxy and its role in inspection decision-making

$\hat{X}$ The empirical distributions of the remaining useful life (RUL) proxy values obtained at inspection epochs are shown in Fig. 11a–c. The RUL proxy is calculated using Equation (4), where is the state estimate derived from noisy measurements. These distributions provide insight into the informational content available to each strategy for adaptive decision-making.

For the fixed-interval policy ($\Delta T, M$) (Fig. 11a), the histogram of RUL proxies is broad, with mass extending across both short and long

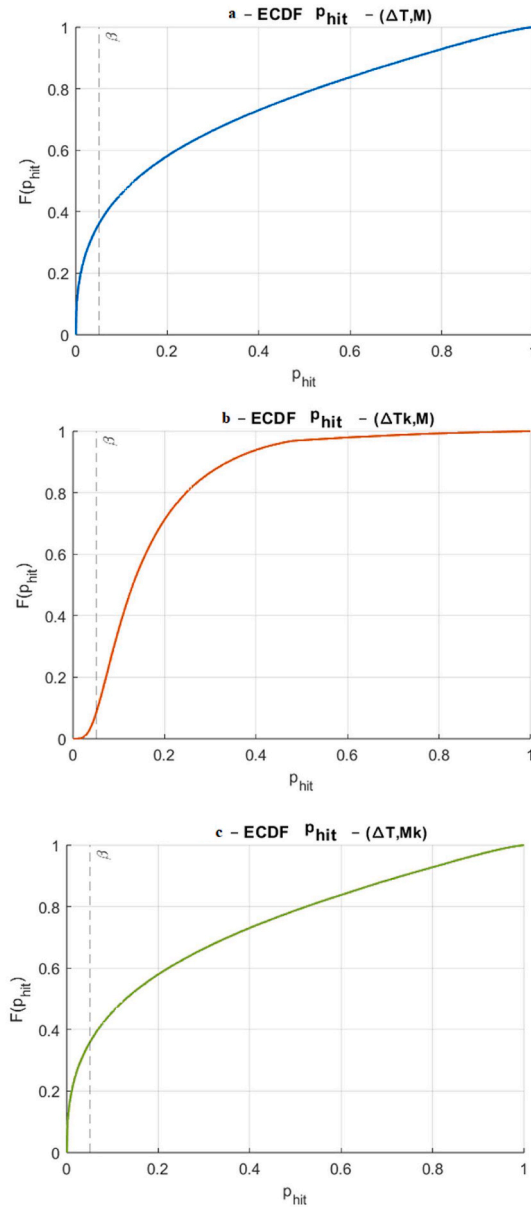


Fig. 15. ECDF p_{hit} for all strategies.

predicted horizons. The distribution reflects the fact that inspections occur at regular intervals regardless of degradation severity. As a result, the RUL estimates frequently capture both early-stage conditions with high residual life and late-stage conditions close to failure. This wide dispersion indicates limited targeting of inspections to periods of heightened risk.

The adaptive-interval policy ($\Delta T_k, M$) (Fig. 11b) yields a distribution that is more concentrated in the region of short-to-moderate RUL values. The clustering around lower RUL proxies is a direct consequence of interval shortening as the system approaches the critical boundary. This concentration implies that inspections are performed more frequently when the system is at elevated risk, increasing the likelihood of timely preventive interventions. The distributional narrowing compared with Fig. 11a demonstrates that adaptive cadence produces more informative inspections by aligning observation epochs with states of greater operational significance.

The adaptive-threshold policy ($\Delta T, M_k$) (Fig. 11c) produces an RUL proxy distribution closely resembling that of the baseline. Since inspections occur at fixed intervals, the distribution spans a wide range of

residual life estimates similar to Fig. 11a. The marginal differences between the two distributions indicate that threshold adaptation does not influence the timing of inspections, and therefore has little effect on the statistical properties of RUL proxies.

The comparison across the three panels shows that interval adaptation alters the informational landscape of inspections by concentrating RUL proxies in ranges where maintenance decisions are more critical. Threshold adaptation leaves the distribution largely unchanged. These results confirm that the adaptive-interval mechanism provides greater potential leverage for risk-responsive decision-making.

5.8. Between-inspection risk

The between-inspection risk, quantified by the probability of crossing the failure threshold before the next scheduled inspection, is summarized in Table 4 and illustrated in Fig. 12a–c, 13a–13c, and 15a–15c.

For the fixed-interval policy ($\Delta T, M$), the mean p_{hit} is 0.2539, with median, 90th percentile, and 99th percentile values of 0.128, 0.734, and 0.958, respectively. The proportion of inspection intervals with $p_{hit} > \beta = 0.05$ reaches 0.641. The histogram in Fig. 2a exhibits a heavy right tail, indicating that many inspection intervals are associated with high failure probabilities. The scatter representation in Fig. 13a confirms that constant inspection intervals do not allow state-dependent modulation of risk. The empirical cumulative distribution function (ECDF) in Fig. 15a demonstrates that most inspection intervals exceed the tolerance limit, reflecting the inability of the policy to constrain risk exposure.

For the adaptive-interval policy ($\Delta T_k, M$), the mean p_{hit} equals 0.2558 and the exceedance fraction is 0.646. The histogram in Fig. 12b preserves the skewed profile observed under the baseline. The scatter in Fig. 13b reveals that shorter intervals correspond to lower p_{hit} , whereas extended intervals are associated with higher risk. The adaptive mechanism therefore redistributes risk across inspection lengths, but the ECDF in Fig. 15b indicates that the overall exceedance probability remains nearly identical to that of the fixed-interval policy. The data confirm that the implemented degree of interval adaptation is insufficient to mitigate high-risk intervals.

For the adaptive-threshold policy ($\Delta T, M_k$), the mean p_{hit} is 0.2543, with an exceedance fraction of 0.643. The histogram in Fig. 12c, scatter in Fig. 13c—and ECDF in Fig. 15c display nearly the same characteristics as the fixed-interval case. Since the inspection cadence is not modified, threshold adaptation exerts no measurable influence on the statistical distribution of between-inspection risk.

The comparative analysis indicates that all three strategies are associated with mean between-inspection failure probabilities around 0.25 and extreme quantiles above 0.95. The adaptive-interval policy modifies the distribution of risk conditional on inspection length but does not alter the aggregate level of vulnerability. The adaptive-threshold policy leaves the risk characteristics virtually unchanged. The persistence of high between-inspection probabilities accounts for the steep decline in survival curves observed in Fig. 8 and explains the inability of all strategies to meet the reliability requirement at the reporting horizon.

5.9. Adaptive threshold statistics

The distribution of preventive maintenance thresholds $L_{m,k}$ is reported in Table 5 and illustrated in Fig. 14a–c. The analysis highlights both the absolute threshold levels and the safety margins relative to the failure boundary L_f .

For the fixed-interval policy ($\Delta T, M$), the threshold remains constant by design. The reported mean value is 2.733, with the 10th and 90th percentiles equal to 2.50 and 3.187, respectively. The corresponding mean safety margin, defined as $L_f - L_{m,k}$, is 2.267. The histogram in Fig. 14a shows a narrow distribution centered on the baseline setting,

indicating the absence of threshold adaptation.

For the adaptive-interval policy ($\Delta T_k, M$), the inspection cadence is adjusted but the threshold remains fixed. The statistics confirm this invariance: the mean threshold is 2.735 with percentiles nearly identical to those of the baseline, and the mean margin equals 2.265. The histogram in Fig. 14b is indistinguishable from Fig. 14a, reflecting the lack of threshold variability under this strategy.

For the adaptive-threshold policy ($\Delta T, M_k$), the threshold is recalculated at each inspection using the estimated remaining useful life. The mean value is 2.734, with a 10th percentile of 2.5 and a 90th percentile of 3.186. The mean safety margin is 2.266. Although the mechanism is designed to introduce adaptivity, the histogram in Fig. 14c shows clustering near the imposed feasibility limits of $L_{\min}=2.5$ and $L_f - \delta L_{\text{safety}}=4.6$. The concentration of outcomes near these bounds indicates that the adaptive rule rarely departs significantly from baseline values.

The statistical evidence demonstrates that the adaptive-threshold policy does not materially alter the distribution of $L_{m,k}$. The similarity of results across all three strategies confirms that threshold adaptation, as implemented here, is effectively constrained by its design bounds and therefore contributes little to improving maintenance performance indicators.

5.10. Penalty-augmented objective

The penalty-augmented objective J_λ incorporates the mean cost rate, the variance of costs, the penalty associated with reliability shortfall at the reporting horizon, and the penalty derived from excessive between-inspection risk. The results are reported in Table 6.

The selection of penalty weights λ_R and λ_{hit} is based on both scaling considerations and engineering priorities. The average cost rate observed in the simulations is on the order of 4–5 units, whereas the reliability shortfall ($R_{\min} - R$) can reach values close to 0.8. To ensure that violations of the reliability constraint dominate the objective function, λ_R is set to 10000, resulting in penalty values on the order of several thousand. This scaling reflects the critical importance of maintaining acceptable reliability levels, particularly in systems where failure consequences are severe.

The between-inspection risk φ_t typically exhibits values around 0.2–0.3, with a significant proportion of intervals exceeding acceptable thresholds. The weight $\lambda_{hit} = 1000$ is selected to produce penalties on the order of hundreds, ensuring that risk is meaningfully penalized while remaining secondary to reliability.

This choice establishes a hierarchical structure in the objective function, where reliability is prioritized over risk, and both are prioritized over cost. The results confirm that the reliability penalty dominates the objective function, indicating that the weighting scheme effectively enforces the intended prioritization. A more systematic exploration of weight selection through multi-objective optimization is identified as a direction for future research.

For the fixed-interval policy ($\Delta T, M$), the mean cost rate is 4.4868 with a variance of 0.768. The reliability at $t=2000$ is 0.165, which implies a shortfall of 0.785 relative to the target requirement of 0.95. The corresponding reliability penalty equals 7850. The fraction of intervals with $p_{hit} > \beta = 0.05$ is 0.641, resulting in a risk penalty of 641. The combined objective value is 8495.4.

The adaptive-interval policy ($\Delta T_k, M$) achieves a slightly lower mean cost rate of 4.4526 and a variance of 0.762. The reliability at the reporting horizon is 0.161, producing a shortfall of 0.789 and a reliability penalty of 7890. The fraction of high-risk intervals is 0.645, yielding a risk penalty of 645. The value of J_λ is 8539.2. Despite the improvement in cost performance, the increased reliability penalty leads to a higher aggregate objective relative to the baseline.

The adaptive-threshold policy ($\Delta T, M_k$) records a mean cost rate of 4.4648 with a variance of 0.709. The reliability level is 0.157, corresponding to a shortfall of 0.793 and a penalty of 7930. The fraction of

high-risk intervals is 0.643, generating a risk penalty of 643. The resulting objective value is 8577.3, the highest among the three strategies.

The comparison of results indicates that the reliability shortfall dominates the augmented objective under all policies. Interval adaptation reduces average costs but fails to improve reliability, which results in a larger J_λ than the baseline. Threshold adaptation has no meaningful effect on cost or risk and produces the least favorable outcome. The penalty structure emphasizes that strategies unable to raise survival probabilities at the reporting horizon cannot achieve competitive values of the augmented objective, regardless of small improvements in cost efficiency.

5.11. Comparative synthesis

The findings presented in Tables 1–6 and Figs. 4–15 offer a comprehensive assessment of the performance of the three maintenance strategies across various operational metrics. Each strategy's influence on degradation trajectories, cost, downtime, action distributions, reliability, and penalty-augmented objectives is distinct, although no strategy achieves the required reliability at the reporting horizon.

The fixed-interval baseline policy ($\Delta T, M$) demonstrates the limitations of a rigid inspection schedule. The degradation trajectories (Fig. 4a) exhibit substantial variability, with high costs and downtime (Figs. 5a and 6a). The PM/CM ratio (1.5832) reflects the lack of optimization in the allocation of preventive and corrective maintenance (Table 3). The survival probability at $t=2000$ is 0.165, a clear indication that the fixed schedule fails to prevent early failures (Table 1). The distribution of between-inspection risk shows a mean $p_{hit}=0.254$ (Table 4), suggesting frequent high-risk intervals. The penalty-augmented objective is 8495.4, driven by the high reliability penalties despite frequent preventive maintenance (Table 6).

The adaptive-interval policy ($\Delta T_k, M$) introduces inspection scheduling based on system degradation, which results in a more favorable cost–downtime profile. The degradation trajectories (Fig. 4b) show that inspections are concentrated around critical states, resulting in reduced mean cost (4.4526) and downtime (15282) relative to the baseline (Table 1). The PM/CM ratio increases to 1.6076 (Table 3), indicating an enhanced balance between preventive and corrective maintenance. However, the survival probability at $t=2000$ remains virtually unchanged at 0.161 (Table 1), indicating that despite improvements in cost and downtime, reliability is not significantly improved. The mean between-inspection risk is 0.256, with a nearly identical exceedance fraction compared to the baseline (Table 4). The value of the penalty-augmented objective increases to 8539.2, highlighting that the improvement in cost is offset by increased reliability penalties (Table 6).

The adaptive-threshold policy ($\Delta T, M_k$) produces results that are similar to those of the baseline. The degradation trajectories (Fig. 4c) show minor advances in intervention timing, but the statistical outcomes are almost identical. The mean cost (4.4648), mean downtime (15317), and PM/CM ratio (1.5876) are nearly the same as those for the baseline (Tables 1–3). The reliability at $t=2000$ is 0.157 (Table 1), the lowest among the strategies. The between-inspection risk distribution (Figs. 12c, 13c and 15c) is nearly identical to the baseline, and the penalty-augmented objective reaches 8577.3, the highest among the strategies (Table 6). This outcome suggests that the threshold adaptation does not significantly impact the risk profile or cost–reliability trade-off.

In comparison, the adaptive-interval policy ($\Delta T_k, M$) produces the most favorable results in terms of cost and downtime while achieving an improved PM/CM ratio. However, this improvement does not translate into a measurable gain in reliability, as evidenced by the near-identical survival probabilities across all strategies. The adaptive-threshold policy produces results almost identical to the baseline, with marginal improvements in cost efficiency but no significant impact on reliability or between-inspection risk.

Across all three strategies, the fixed-interval baseline yields the

highest reliability relative to the adaptive-interval and adaptive-threshold policies, but it is the least efficient in terms of cost and downtime. The inability of any strategy to reduce between-inspection risk is a key factor contributing to the lack of compliance with the reliability requirement, as reflected in the consistently low survival probabilities and high penalty-augmented objectives.

6. Engineering implications

The findings of this study provide important insights for the design and implementation of maintenance strategies in various industrial settings. While the adaptive-interval policy demonstrated improvements in cost and downtime performance, no strategy achieved the required reliability at the reporting horizon. This outcome highlights the challenges of mitigating between-inspection risk and emphasizes the need for more advanced maintenance frameworks in industrial environments.

In critical industries such as aerospace, power generation, and heavy manufacturing, system reliability is often a primary concern. The results indicate that while adaptive-interval strategies improve maintenance efficiency, they are insufficient to guarantee high reliability. In these industries, the implementation of predictive maintenance systems that incorporate real-time monitoring and machine learning algorithms could significantly reduce between-inspection risk. These systems would allow for more precise interventions, ensuring that potential failures are addressed before they occur, thus improving reliability while maintaining cost-effectiveness.

The adaptive-threshold policy demonstrated limited impact due to constraints on threshold adjustments, which were often confined by predefined bounds. In industries with highly variable operating conditions, such as oil and gas and mining, threshold-based policies may be effective but require greater flexibility in inspection intervals. The adaptability of both thresholds and inspection timings would allow for more optimal reliability outcomes, enabling interventions that are both timely and cost-efficient.

Furthermore, the results derived from the penalty-augmented objective (Table 6) underscore the importance of reliability in industries where failure consequences are critical, such as aviation, nuclear power, and rail transportation. These sectors demand maintenance strategies that minimize failure risks while balancing operational costs. For such industries, the integration of predictive maintenance technologies and condition-based monitoring systems is essential. These technologies allow for the dynamic adjustment of maintenance schedules based on real-time asset conditions, leading to lower maintenance costs while preserving system reliability.

The analysis of between-inspection risk, particularly the p_{hit} values, reinforces the need for integrated risk management frameworks in industrial settings. These frameworks should account for both degradation and failure risks while also evaluating the broader operational impact of maintenance decisions. In manufacturing industries, for example, unplanned downtime resulting from inadequate preventive maintenance can have significant financial consequences. Therefore, the adoption of predictive maintenance that adjusts based on real-time data could optimize maintenance schedules, balancing the need for cost reduction with system reliability.

While the findings suggest that no policy currently achieves both low costs and high reliability, they also indicate the potential for substantial improvement through the development of more sophisticated maintenance strategies. The integration of advanced diagnostics and machine learning models will improve predictive capabilities, leading to more precise and timely maintenance interventions. Such strategies will likely result in improved operational efficiency, enhanced reliability, and reduced maintenance costs.

From a practical implementation perspective, the computational requirements of the proposed maintenance strategies remain moderate. The degradation model is based on a Wiener process, which can be simulated using simple numerical updates, and the adaptive decision

rules involve only basic algebraic operations. As a result, the adaptive-interval policy does not introduce significant computational overhead compared to the fixed-interval approach.

In real-world applications, the adaptive-interval strategy can be integrated into IoT-based condition monitoring systems, where sensor data is continuously collected to estimate the degradation state of the system. The remaining useful life (RUL), used to adjust inspection intervals, can be computed efficiently in real time, making the approach suitable for online implementation.

However, practical deployment requires reliable sensing infrastructure, appropriate data acquisition frequency, and robust estimation of degradation states under measurement uncertainty. In addition, communication and processing constraints in industrial IoT environments must be considered when implementing adaptive maintenance policies. Despite these challenges, the simplicity of the proposed adaptive mechanisms makes them compatible with existing monitoring systems and highlights their potential for real-world applications.

In conclusion, industries where both availability and safety are critical must prioritize a comprehensive approach to maintenance. This approach should combine predictive analytics, real-time monitoring, and adaptive maintenance scheduling to meet the dual objectives of minimizing costs and maximizing reliability. The findings of this study underscore the importance of continuously evolving maintenance strategies to meet the dynamic demands of industrial operations.

The present study is based on simulated degradation data generated using a stochastic Wiener process model. While this approach provides a controlled and flexible environment for evaluating maintenance strategies, it does not fully capture all complexities of real-world systems. In practice, degradation behavior may be influenced by additional factors such as environmental variability, measurement noise, and operational uncertainties.

Nevertheless, simulation-based analysis remains a widely accepted approach for the development and comparison of maintenance policies, as it allows systematic exploration of strategy performance under controlled conditions. The modeling assumptions adopted in this work are consistent with established practices in condition-based maintenance and provide meaningful insights into the structural behavior of adaptive strategies.

Future work should focus on validating the proposed framework using real degradation or maintenance datasets, as well as integrating real-time monitoring data from industrial systems. Such validation would enable the assessment of model accuracy under practical conditions and further enhance the applicability of the proposed approach.

7. Conclusion conclusions and perspectives

This study presents a detailed evaluation of three maintenance strategies in the context of adaptive condition-based maintenance (CBM). Using Monte Carlo simulations, we have compared the performance of the fixed-interval policy ($\Delta T, M$), the adaptive-interval policy ($\Delta T_k, M$), and the adaptive-threshold policy ($\Delta T, M_k$). The evaluation focused on key performance metrics including cost, downtime, reliability, and the penalty-augmented objective. The results indicate that while improvements were observed in cost and downtime under the adaptive strategies, none of the policies achieved the desired reliability at the reporting horizon, highlighting the need for more robust approaches to manage between-inspection risks.

The fixed-interval policy exhibited substantial shortcomings, particularly in cost and downtime. Its rigid inspection intervals failed to prevent early failures, resulting in significant reliability shortfalls. In contrast, the adaptive-interval policy improved cost efficiency and reduced downtime, but it did not lead to a significant increase in reliability, as it did not sufficiently address the risk of failures occurring between inspections. The adaptive-threshold policy, although incorporating threshold adjustments, showed negligible improvement in both cost and reliability. The constraints imposed by fixed thresholds limited

the potential of this strategy to adapt effectively to varying system conditions.

The findings underscore the importance of more advanced, data-driven maintenance strategies. Real-time monitoring systems, predictive maintenance models, and machine learning techniques offer significant potential to reduce the risk of failures between inspections while improving cost efficiency. These technologies can enable more targeted, dynamic interventions, based on the real-time condition of the system. Future research should focus on developing adaptive maintenance strategies that integrate both interval adjustments and threshold modifications, potentially in combination with model-based approaches such as digital twins.

Further advancements in condition monitoring and predictive analytics are required to enhance maintenance decision-making. Future studies should explore the potential of multi-objective optimization methods to address the competing demands of cost reduction, reliability enhancement, and downtime minimization. These methods could enable the development of strategies that more effectively balance the trade-offs between different maintenance objectives.

The limitations observed in this study point to the need for a more holistic view of system health, incorporating real-time data and sophisticated diagnostics. As industries move towards more integrated, digital maintenance frameworks, further work is needed to develop more precise and flexible maintenance schedules. These schedules should better reflect the evolving state of the system, allowing for interventions that are both timely and cost-effective.

In conclusion, this study highlights the need for ongoing innovation in maintenance strategies. The combination of real-time condition monitoring, predictive modeling, and adaptive decision-making will be crucial for achieving optimal maintenance outcomes in complex industrial systems, particularly those where reliability and operational efficiency are critical.

CRedit authorship contribution statement

Khamiss Cheikh: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **EL Mostapha Boudi:** Visualization, Validation, Supervision.

Declaration of competing interest

Khamiss Cheikh and EL Mostapha Boudi declare that they have no conflict of interest or financial conflicts to disclose. We affirm our commitment to upholding the highest standards of ethical conduct in scientific research. We declare that the work presented in this submission is original and has not been published elsewhere. All named authors have significantly contributed to the research and agree to take responsibility for its contents.

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Data availability

Data will be made available on request.

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